Definitions

- Statements and Statement Operations ✓
  - A statement has a *definite* truth value
  - Algebraic equations are statements if the variables are quantified (e.g. $\forall x \in \mathbb{R}, x + 1 = 2$, which is definitely false)
- Sets and Set Operations ✓
  - $(A \cup B)^c = A^c \cap B^c$
  - $(A \cap B)^c = A^c \cup B^c$
  - $A \setminus B = (A \cap B^c)$
  - $\cup, \cap$ are commutative.
  - $\cup, \cap$ are associative.
  - $\cup, \cap$ distribute like multiplication; $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
  - $A \cup \emptyset = A$
  - $A \cap U = A$
  - $A \cup U = U$
  - $A \cap \emptyset = \emptyset$
  - $A \cup A = A$
  - $A \cap A = A$
  - $A \cup A^c = U$
  - $A \cap A^c = \emptyset$
  - $(A^c)^c = A$
- Cardinality of Finite Sets ✓
  - $|A \times B| = |A||B|$
  - $|P(A)| = 2^{|A|}$
- Power Set ✓
  - $P(A) = \{X \mid X \subseteq A\}$
  - Don't forget the empty set $\emptyset$!!!!!!!!!!!!!!!!!!!!!
- Divisibility ✓
  - $a|b \iff \exists m \in \mathbb{Z}, b = ma$
- 4 Types of Induction ✓
  - PMI: $P(1) \land [P(n) \implies P(n+1)] \implies \forall m \in \mathbb{N}, P(m)$
  - PSI: $P(1) \land [\bigwedge_{k=1}^{n} P(k) \implies P(n+1)] \implies \forall m \in \mathbb{N}, P(m)$
  - WOP: $\forall S \subseteq \mathbb{N}, S \neq \emptyset, S \text{ has a least element.}$
  - DCC:

    - No infinite descending chains exist in the naturals.
    - Every descending chain in the naturals must reach 0 or a constant;
      $n_1 \geq n_2 \geq \ldots \implies (\exists k \in \mathbb{N}, n_k = 0 \text{ or } n_k = n_{k+1} = n_{k+2} = \ldots)$
- Relations & Properties ✓
    - A relation R is a subset of the Cartesian product of two sets; $R \subseteq \{(a, b) \in A \times B\}$.
    - A relation R on set A is a subset of the Cartesian product of A and itself;
      $R = \{(a_1, a_2) \in A \times A\}$.
    - If two elements $a_1, a_2 \in A$ are related by relation R (i.e. $\exists (a_1, a_2) \in R$), we write $a_1 R a_2$.
    - For relation R on set A and $a,b,c \in A$:
        - R is reflexive $\iff \forall a \in A, aRa$; every element in A relates to itself by R. THIS IS THE ONLY PROPERTY WITH A UNIVERSAL CONDITION!!!
        - R is symmetric $\iff (aRa \implies bRa)$; if a is related to b, then b must be related to a by R.
        - R is antisymmetric $\iff (aRb \wedge bRa \implies a = b)$; if a and b are symmetrically related, they must be the same element.
        - R is transitive $\iff (aRb \wedge bRc \implies aRc)$; if a is related to b and b is related to c, a must be related to c by R.
- Partial Orders ✓
    - Reflexive, Antisymmetric, Transitive (RAT)
    - poset: (S, R)
    - For poset $(S, R), x, y \in S$:

$$x \text{ is the greatest element} \iff \forall y, x \geq y$$

$$x \text{ is the least element} \iff \forall y, x \leq y$$

$$x \text{ is a maximal element} \iff \nexists y, \text{ s.t. } y > x.$$

$$x \text{ is a minimal element} \iff \nexists y, \text{ s.t. } y < x.$$

      NOTE: For x to be a greatest/least element it MUST be comparable to EVERY ELEMENT IN THE POSET!!!!!!!!!!!!!!!!!!
      NOTE: It is vacuously true that if an element $x \in S$ is not related to any other element, then it is both minimal and maximal by those definitions.
      NOTE: It is true that if a poset has strictly greater than 1 element, and there is at least one isolated element in that poset, then that poset has no greatest or least element.
    - x is greatest $\implies x$ is maximal
    - x is least $\implies x$ is minimal
- Total/Linear Orders ✓
    - Partial Order + Trichotomy Law (total comparability)
    - Trichotomy Law: For $x, y \in S, x < y, x = y, \text{ or } x > y$ but only one.

- For a toset $(S, R), x, y \in S$:

$$\begin{aligned} x \text{ is greatest} &\implies x \text{ is the unique maximal} \\ x \text{ is least} &\implies x \text{ is the unique minimal} \end{aligned}$$

Problems

- Completing Truth Tables ✓
- Writing Statements with Quantifiers ✓
- Negating Statements ✓
  - $\neg\forall x = \exists x$
  - $\neg\exists x = \forall x$
  - $\neg(a \leq b) = a > b$
  - $\neg(a \geq b) = a < b$
- Intersections and Unions of Sets -> Arbitrary $\bigcap$ & $\bigcup$ of Indexed Sets?
  - For a given sequence of indexed sets, there are two mechanical workflows depending on how the sets are indexed:
    - $\{A_n\}_{n=1}^{\infty}$ is usually written interval notation (open, closed, or half-open) in HW 2.
      1. If the sequence is nested and decreasing, $\bigcup A_n = A_1$, or $A_{\text{first index}}$.
      2. If the sequence is nested and increasing, $\bigcap A_n = A_{\text{first index}}$.
      3. Use the prior steps if applicable, then follow these steps if A_n is nested:
         For $\bigcup A_n$, take the limit as $n \to \infty$. This gives you an interval with endpoints $(\lim f(n), \lim g(n))$. If A_n is closed or half-closed, check that the limit boundary value is actually in the interval: $f(n) < \lim f(n) < g(n)$? If not, change the closed bracket to an open bracket.
         For $\bigcap A_n$, take the limit as $n \to \infty$. If A_n is open, this will likely result in a empty interval (a, a). If $f(n) < a < g(n)$, then $\bigcap A_n = a$, else it is the empty set. If A_n is closed, no need to test the boundary value.
      4. If A_n is not nested:
         For $\bigcup A_n$, write out the first few terms of the union and infer the tail-end behavior of the endpoints. For $\bigcap A_n$, write out the first few terms of the intersection and either infer the tail-end behavior of the endpoints directly or evaluate the first few intersections to see if you get $\emptyset$. If you ever get $\emptyset$, then the entire intersection must be $\emptyset$.
    - $\{A_k\}_{k\in\Lambda}$ is followed by a definition of a sequence of sets, not intervals in HW 2.
      1. For $\bigcup A_k$, write out the first few terms of the union and infer what the resultant set is. Try to think about what indexing the set by Λ really means when performing the union over all of the sets. For example, in problem 8c), we union over all $k \in \mathbb{Z}$ of $\{k, k+1, \ldots\}$, so by inspection this should just be $\mathbb{Z}$.
         For $\bigcap A_k$, write out the first few terms of the intersection and infer what the

resultant set is. For the same problem, we notice that, for each subsequent increment in k, we lose the first element from the preceding set, so intersecting over all of k gives us the empty set.

    - Double Arbitrary ⋂ & ⋃ of Indexed Sets ? <- On the exam???
- Proving Set Inclusions
    - Let $x \in S$, use the property defining $x \in S$ to show that $x \in T$, conclude $S \subseteq T$.
    - $S = T \iff (S \subseteq T) \land (T \subseteq S)$
- Proof by Contrapositive ✓
    - Proof by contrapositive in an set inclusion proof
        - Proving set inclusion requires proving $S \subseteq T$ and $T \subseteq S$, which are two implicit conditional statements: $\forall s, s \in S \implies s \in T$ and $\forall t, t \in T \implies t \in S$, respectively. So invoke the contrapositive for each conditional you are trying to prove, and prove that contrapositive directly. Recall: $P \implies Q \equiv \neg Q \implies \neg P$.
    - Proof by contrapositive in a uniqueness proof
        - A uniqueness proof follows the form: $P(x) \land P(y) \implies x = y$, so naturally there is a way to prove this using the contrapositive by proving $x \neq y \implies \neg P(x) \lor \neg P(y)$.
- Standard Induction Proofs
    - Sum = formula proof: reindex sum, invoke IH, show P(n+1)
    - $a|f(n)$ proof: $P(1)$->assume $a|f(n)$->write out $f(n+1)$->manipulate into $f(n) +$ something->can split divides operator by linearity property to focus on $a|\text{something}$->if you can, reduce $a|\text{something}$ into $2|n(n+1)$, which is true because $n(n+1)$ is guaranteed to be a multiple of an even number, which is divisible by 2.